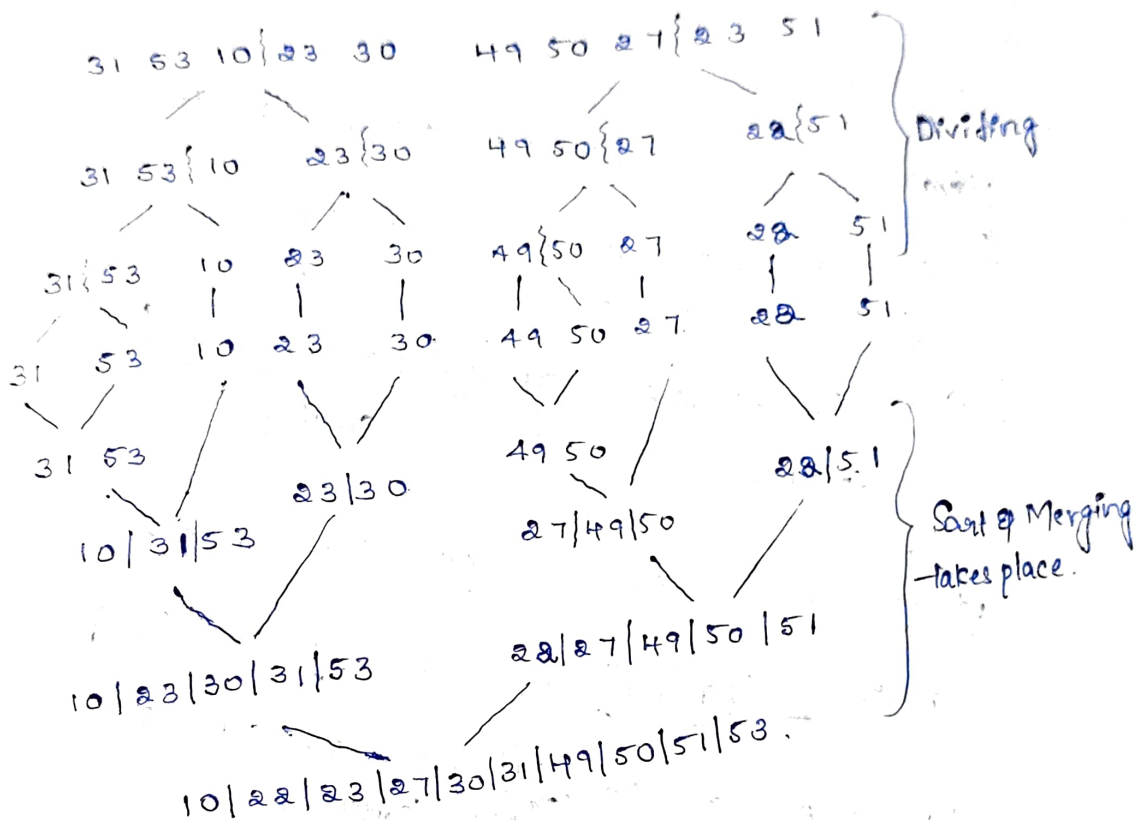


(1) 31 53 10 | 23 30 | 49 50 27 | 22 51
 left mid mid+1 right

Divide by using mid using Divide.



merge Sort (arr, low, mid, high) {

l → mid

mid+1 → right

temp

arr = [];

left = low;

right = mid+1;

while (left <= mid && right <= ^{high}mid+1) {

if (arr[left] <= mid && arr[right] <= ^{high}mid+1) {

(or) if (arr[left] <= arr[right]) {

temp.arr[left]

left++;

else

temp.arr[right].

right++;

}

is on left ✓

while (left <= mid) {

temp.arr[left]

left;

}

is on right ✓

while (right <= high) {

temp.arr[right]

right;

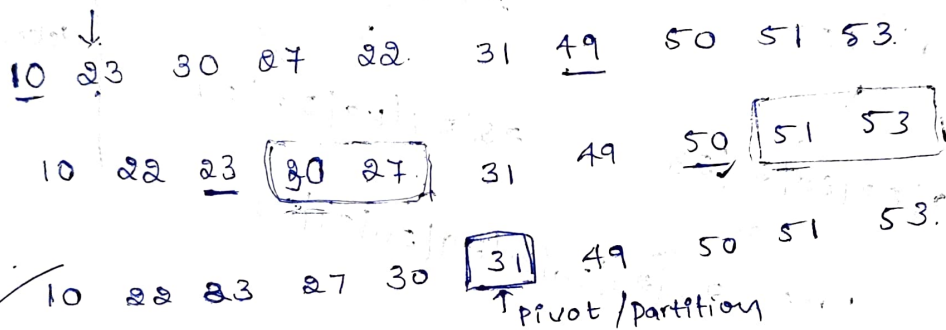
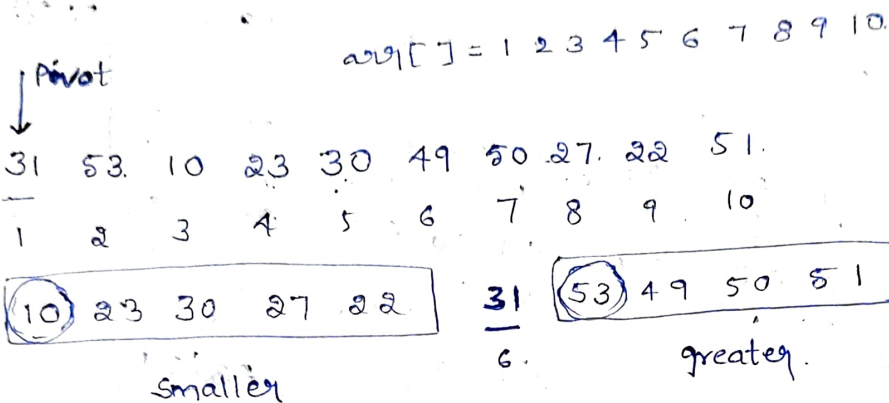
}

for (i = high $\rightarrow$ low) {

arr[i] = temp[i - low];

}

(a)



qs(low, partition - 1)

qs(high, partition + 1)

qs(arr, low, high) {

if (low < high) {

partition = qs(arr, low, high);

qs(arr, low, partition - 1);

qs(arr, partition + 1, high);

}

}